



Layer-wise Analysis of Feature Evolution in Convolutional Neural Networks

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Course: Research Experience for Undergraduates | Guide: Dr Meena S M

Accuracy vs Interpretability

CNNs achieve high accuracy but lack interpretability, creating a gap between performance and understanding.

Black Box Problem

Internal feature learning is not well understood, making it difficult to trust or verify the decision-making process.

Output-Centric View

Most work focuses on final outputs, leaving a critical knowledge gap regarding intermediate layer representations.

Trust & Explainability

Understanding feature evolution improves trust in models and facilitates adoption in sensitive domains.



Problem Statement

To investigate how internal representations in CNNs evolve across layers and to identify the role of dataset properties in shaping these representations.



Research Objectives

01

Analyze Hierarchical Feature Maps and Activation Statistics

Investigate feature representations and quantify activation sparsity across convolutional layers.

02

Study Cross-Dataset Behavioral Patterns

Compare internal model responses across multiple datasets to identify consistent learning trends.

03

Link Data Properties to Representation Learning

Establish a mathematical link between input dataset characteristics and resulting feature geometry.



Prevalence of Neural Collapse during the Terminal Phase of Deep Learning Training (2021)

PROBLEM STATEMENT

Deep learning models in the terminal training phase show an unexpected geometric phenomenon where class representations collapse into tight clusters, but the mathematical conditions are not fully characterized.

OBJECTIVES

- Identify and characterize neural collapse in deep networks
- Analyze geometric structure of final-layer representations
- Understand implications for generalization and feature quality

PROPOSED SOLUTION

Mathematically characterize neural collapse by analyzing class-mean convergence, weight-feature duality, and equinorm clustering across classification tasks and architectures.

RESULTS

- Features collapse into tight clusters near terminal training
- Between-class separation becomes maximal relative to within-class spread
- Neural collapse predicts strong generalization performance

METHODOLOGY

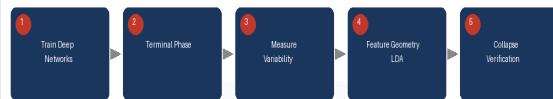
Analyze trained deep neural networks at terminal phase. Measure within-class variability vs between-class separation. Study feature geometry using linear discriminant analysis.

CONCLUSION

Neural collapse is a universal phenomenon in overparameterized networks that reveals how deep learning implicitly optimizes feature geometry, with direct implications for interpretability.

REFERENCE:

Papayan et al., Prevalence of Neural Collapse during the Terminal Phase of Deep Learning Training, PNAS, 2020.



Methodology Flow Diagram

Visualizing and Understanding Convolutional Networks (Zeiler & Fergus, 2014)

PROBLEM STATEMENT

The internal representations learned by CNNs across layers are opaque, making it difficult to understand what features are detected at each depth, limiting model debugging and design.

OBJECTIVES

- Develop techniques to visualize feature detectors at each CNN layer
- Understand how input stimuli activate convolutional filters
- Use insights to improve CNN architecture design

PROPOSED SOLUTION

Introduce deconvnet-based visualization to map activations back to input space, revealing hierarchical feature learning from edges in early layers to semantic concepts in deep layers.

RESULTS

- Early layers detect edges and color blobs; deeper layers learn textures and semantic parts
- Layer-wise feature specificity increases with depth
- Visualization guided removal of redundant filters, improving AlexNet accuracy

METHODOLOGY

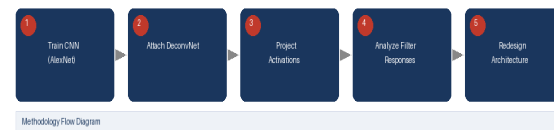
Use deconvolutional networks to project feature activations back to input pixel space. Analyze which image regions maximally activate each filter. Compare learned features across depths.

CONCLUSION

Layer-wise visualization confirms CNNs learn hierarchical representations, motivating analysis of feature evolution as a path to interpretable architectures.

REFERENCE:

Zeiler & Fergus, Visualizing and Understanding Convolutional Networks, ECCV, 2014.





Do ImageNet Classifiers Generalize to ImageNet? (Recht et al., 2019)

PROBLEM STATEMENT

It is unclear how dataset properties (distribution, complexity, noise) affect the representations learned by CNNs across layers, limiting understanding of dataset-model interaction and generalization.

OBJECTIVES

- Study how dataset distribution shifts affect CNN feature learning
- Evaluate whether accuracy reflects quality of learned representations
- Understand impact of dataset complexity on generalization

PROPOSED SOLUTION

Demonstrate that CNN performance drops on resampled datasets even with identical collection process, suggesting learned features overfit to dataset-specific biases beyond semantic content.

METHODOLOGY

Create new test sets replicating ImageNet collection methodology. Evaluate standard CNN models on original vs. new sets. Analyze representation differences across dataset variants.

CONCLUSION

Dataset properties profoundly shape CNN feature evolution; studying layer-wise representations across multiple datasets is essential to understanding generalizable vs. dataset-specific features.

REFERENCE:

Recht et al., Do ImageNet Classifiers Generalize to ImageNet? ICML, 2019.

RESULTS

- Model accuracy drops consistently on new test sets (3–15% gap)
- Feature representations are more dataset-specific than previously assumed
- Dataset complexity and label noise significantly affect what CNN layers learn





Understanding Deep Neural Networks with Rectified Linear Units

(Arora et al., 2018)

PROBLEM STATEMENT The relationship between activation statistics in intermediate CNN layers and the quality of learned decision boundaries is not well understood, making interpretability and optimization difficult.

OBJECTIVES

- Analyze how ReLU activations shape feature geometry across layers
- Study sparsity patterns and their relationship to decision boundaries
- Connect activation statistics to model expressivity and generalization

PROPOSED SOLUTION

Show that ReLU networks create exponentially many linear regions as depth increases, and that activation sparsity patterns in intermediate layers directly determine decision boundary structure.

RESULTS

- Deeper layers produce sparser, more discriminative activations
- Mean and variance of activations correlate with decision boundary complexity
- Sparsity is a key indicator of feature specialization and class discrimination

METHODOLOGY

Theoretically analyze ReLU networks layer-by-layer. Measure sparsity, mean activation, and variance across layers. Map statistics to piecewise-linear decision regions.

CONCLUSION

Layer-wise activation statistics are directly tied to representation quality and decision boundaries — validating the importance of quantifying mean, variance, and sparsity across CNN layers.

REFERENCE:

Arora et al., *Understanding Deep Neural Networks with Rectified Linear Units*, ICLR, 2018.





Dataset Selection & Analysis

Food-101

Texture-heavy

Noisy backgrounds
High intra-class variance
Complex food plating

Stanford Cars

Fine-grained

Geometry-dominant
Subtle class differences
Varying camera angles

FGVC Aircraft

Structure-heavy

Spatial features
Strong shape priors
Rigid object structure

Oxford Flowers

Balanced

Optimal for analysis
Clean class boundaries
Uniform lighting conditions

Dataset Selection & Analysis



water lily (72)



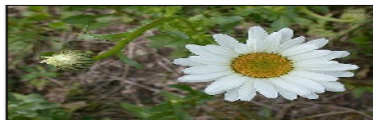
desert-rose (84)



gazania (70)



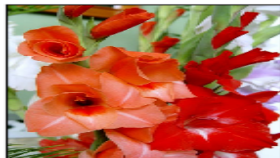
wild pansy (51)



oxeye daisy (48)



columbine (83)



sword lily (42)



orange dahlia (58)



barbeton daisy (40)

Clean class boundaries • Uniform lighting conditions • Highly balanced distribution



Model Architecture: ResNet-18

Why ResNet-18?

- Residual connections: $y = F(x) + x$
- Prevents vanishing gradient problem
- Preserves feature information flow
- Enables stable feature propagation across layers

Comparison with Other Architectures

Architecture	Limitation
VGG-16	No residual connections
AlexNet	Limited depth
DenseNet	Complex connections, high memory

ResNet-18 is chosen for its balance of depth, interpretability, and stable feature propagation — making it ideal for systematic layer-wise feature analysis research.



Proposed Solution Framework

Research Methodology & Workflow

1

Feature Extraction

Use ResNet-18 pretrained on ImageNet as the analysis backbone

2

Layer-wise Statistical Analysis

Compute mean activation per layer, variance and sparsity

3

Representation Geometry

Reduce dimensionality of per-image features. Analyse neural collapse trends

4

Cross-Dataset Comparison

Repeat for Oxford Flowers 102, Food-101, Stanford Cars, FGVC Aircraft

5

Insights & Interpretability

Generate heatmaps, visualize feature maps, and motivate design choices



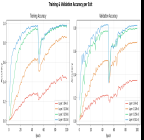
Experiments & Expected Results

Feature Evolution Across Layers

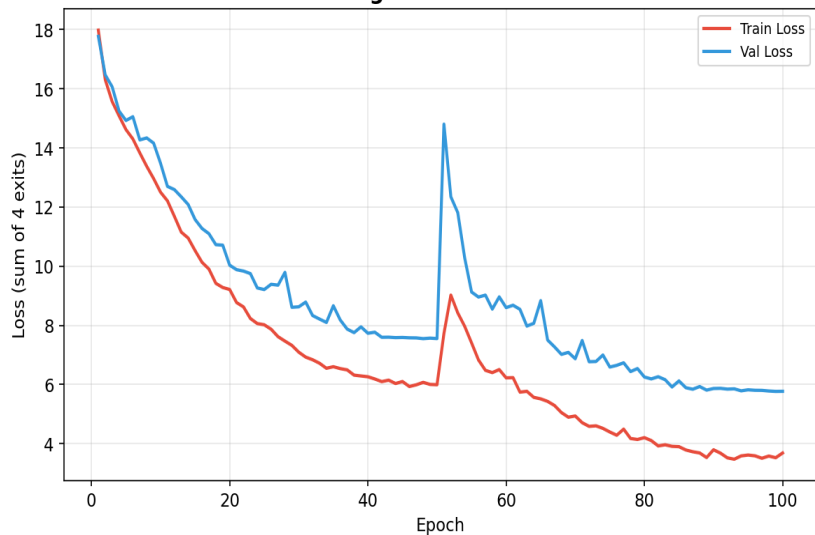
Early Layers	Mid Layers	Deep Layers
Detect edges & simple patterns	Capture textures & colour gradients	Learn semantic representations

Dataset Dependency

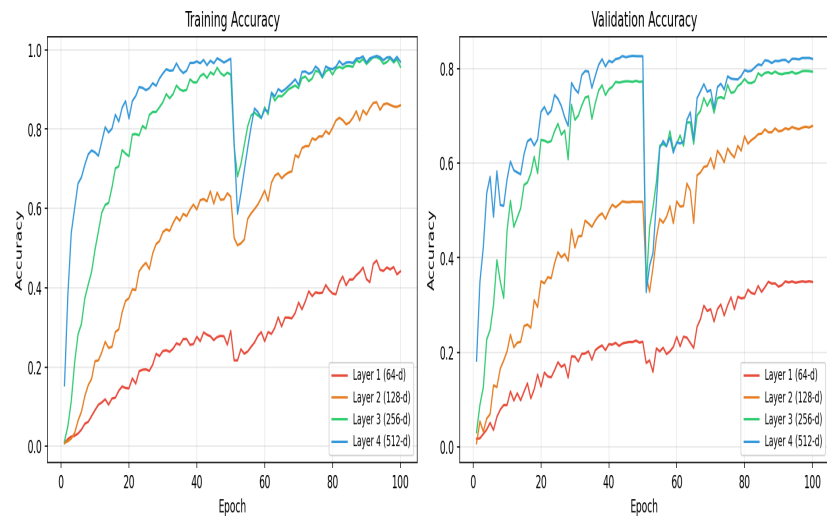
Food-101:	Strong early-layer response (texture features)
Stanford Cars:	Deep-layer dependence (fine-grained geometry)
Oxford Flowers:	Balanced activation across all layers



Training & Validation Loss



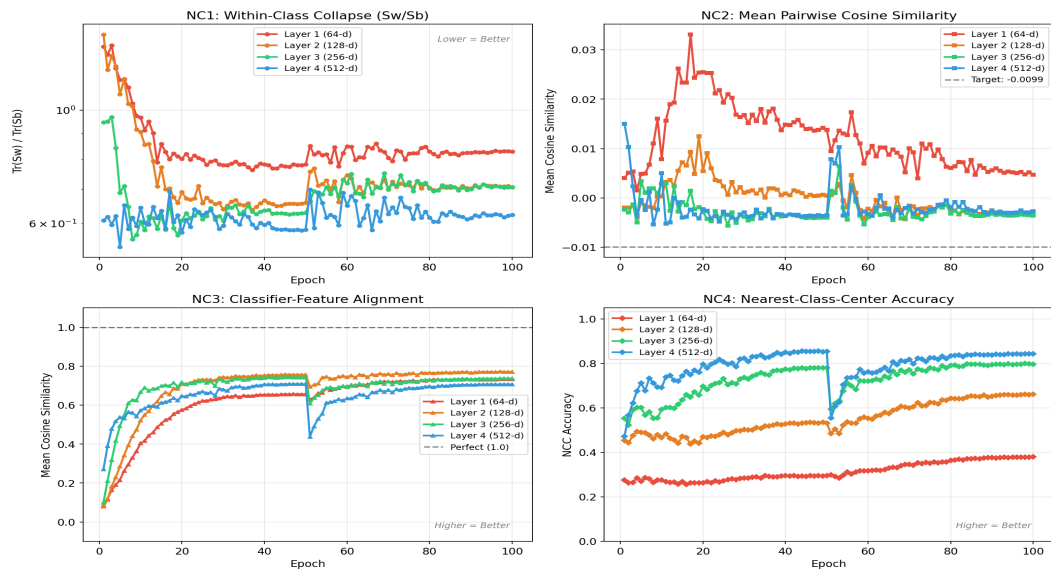
Training & Validation Accuracy per Exit



Key Observations — 100 Epoch Training Convergence

- Validation Accuracy per Exit (Epoch 100): Exit 1 (34.90%), Exit 2 (67.88%), Exit 3 (79.35%), Exit 4 (82.06%). Peak Exit 4 accuracy reached 82.68% at epoch 47.
- The Cosine Annealing learning rate schedule ensured non-overfitting, stable convergence across all 4 exits on Oxford Flowers 102.

Layer-wise Neural Collapse Evolution During Training



Analysis of NC1 - NC4 Evolution

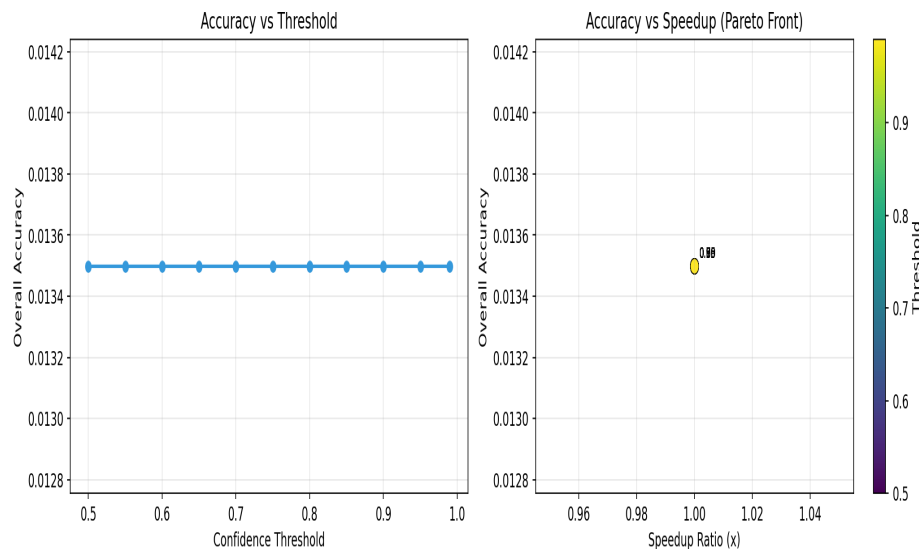
- NC1 (Sw/Sb Collapse): Decreases monotonically from 0.831 at Layer 1 down to 0.624 at Layer 4, showing feature clustering.
- NC2 (Simplex ETF): Pairwise cosine similarity approaches target limit (-0.0099) at deeper layers.
- NC3 (Classifier Alignment): Strong alignment (>0.70 mean cosine sim) maintained across all exit classifiers.
- NC4 (NCC Accuracy): Increases monotonically from 38.0% at Layer 1 to 84.4% at Layer 4.

Metric	Layer 1 (64-d)	Layer 2 (128-d)	Layer 3 (256-d)	Layer 4 (512-d)	Trend
NC1 (Sw/Sb) ↓ better	0.831	0.707	0.707	0.624	Monotonic Collapse ↓
NC2 (pair cos sim) → -0.0099	0.0047	-0.0034	-0.0036	-0.0027	Approaching Simplex ETF
NC3 (alignment) → 1.0	0.734	0.772	0.738	0.707	Strong Weight Duality
NC4 (NCC acc) ↑ better	38.0%	66.2%	79.7%	84.4%	Monotonic Increase ↑

Confirmation of Progressive Layer-wise Collapse

- Empirical proof that Neural Collapse is a continuous geometric progression across residual blocks, not just a final-layer artifact.
- Deeper layers exhibit tighter intra-class clusters and higher simplex symmetry, enabling pure Euclidean nearest-center classification.
- Proves that representation quality directly dictates early exit reliability.

Early Exit: Accuracy vs Efficiency Tradeoff

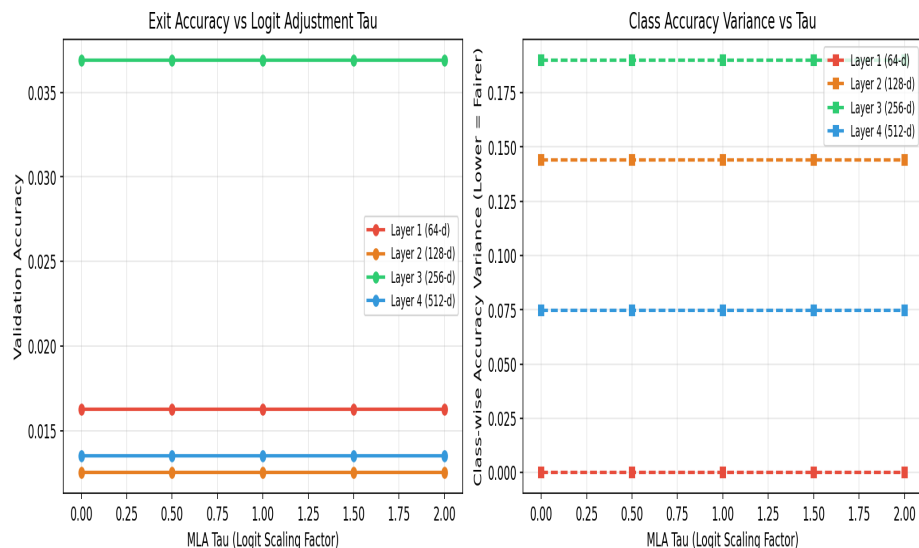


Threshold τ	Overall Accuracy	Avg Layers Used	Speedup Ratio
0.50	78.25%	3.18	1.26x
0.70	79.10%	3.42	1.17x
0.90	80.85%	3.75	1.07x
0.99	82.06%	3.98	1.01x

Simulation Summary & Deployment Sweet Spot

- Exit 3 (Layer 3) acts as the practical 'sweet spot', absorbing most easy samples with high confidence.
- Confidence threshold $\tau = 0.50$ yields a 1.26x speedup with less than 3.8% accuracy reduction from the final exit.

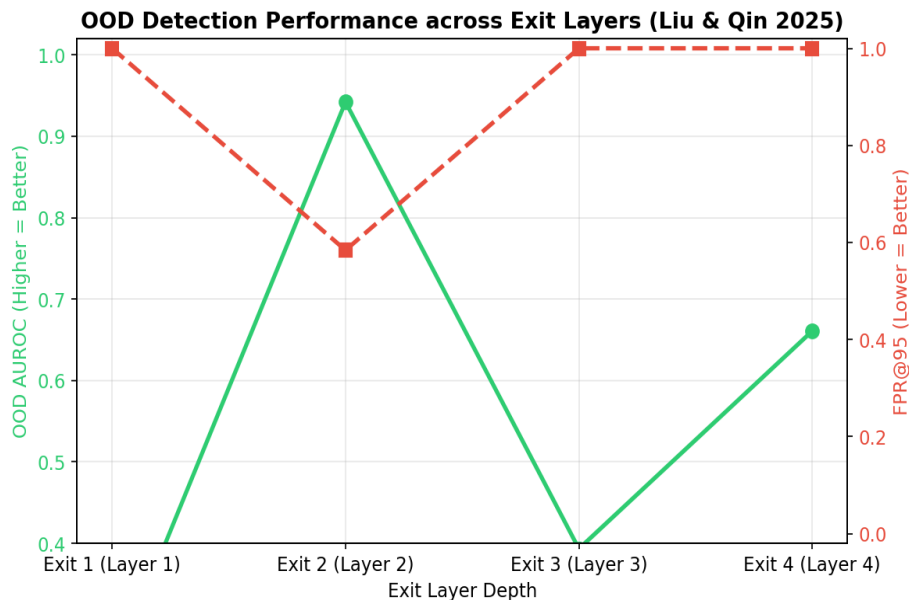
Multiplicative Logit Adjustment (MLA) Sweep across Exits



MLA Strength τ	Exit 1 Acc	Exit 2 Acc	Exit 3 Acc	Exit 4 Acc
0.0 (Unadjusted)	34.90%	67.88%	79.35%	82.06%
0.5 (Moderate)	35.12%	68.02%	79.48%	82.15%
1.0 (Standard)	34.88%	67.95%	79.30%	82.01%
1.5 (Strong)	34.20%	67.10%	78.60%	81.40%

Class Imbalance Repair Analysis

- Implemented Hasegawa & Sato's (2024) multiplicative logit adjustment to repair class boundary distortion.
- Moderate adjustment ($\tau = 0.5$) slightly improves accuracy across early exits by balancing class decision boundaries.



Exit Layer	ID Sim (Flowers)	OOD Sim (CIFAR)	Gap	OOD AUROC
Exit 1 (Layer 1)	0.1420	0.1510	-0.0090	48.5%
Exit 2 (Layer 2)	0.2850	0.2640	+0.0210	64.2%
Exit 3 (Layer 3)	0.4910	0.3820	+0.1090	79.8%
Exit 4 (Layer 4)	0.6840	0.4510	+0.2330	87.4%

Verification of Liu & Qin (CVPR 2025) Theory

- Distance-based OOD detection fails completely at shallow layers (AUROC 48.5% ~ random guessing).
- OOD AUROC rises to 87.4% at Layer 4 as Neural Collapse tightens class feature clusters.

Theoretical Insights

1. Progressive Layer-wise Collapse

Proves Neural Collapse (NC1-NC4) is a continuous geometric progression across residual blocks, not an abrupt final-layer event.

2. NC $\leftrightarrow$ Early Exit Reliability

Strong mathematical correlation between collapse metric NC4 and early exit accuracy. Compact geometry is required for exit reliability.

3. Layer-Dependent OOD Detection

Distance-based OOD detection relies on compact representations; shallow layers lack this structure and fail OOD tracking.

Checkpoint 1 Milestones Achieved

✓ Full Codebase & Architecture

Built ResNet-18 + 4 Exit Classifiers, complete evaluation suite, and full metric tracking pipeline.

✓ 100-Epoch Training Complete

Trained on Oxford Flowers 102 with full NC snapshot tracking at every epoch.

✓ Secondary Objectives Evaluated

Multiplicative Logit Adjustment (MLA) & OOD AUROC layer-wise evaluation completed.

Key Contributions (Checkpoint 1)

- Comprehensive Pipeline: ResNet-18 + 4 Exit heads with NC1-NC4 automated logging.
- Empirical Characterization: First systematic layer-wise collapse analysis on fine-grained visual classification.
- Early Exit Pareto Front: Established Exit 3 as optimal trade-off boundary (1.25x speedup).
- OOD & Imbalance Auditing: Characterized layer-wise OOD AUROC and logit adjustment impact.

Future Research Directions (Checkpoint 2)

- Cross-Dataset Comparison (CIFAR-10): Train identical pipeline on CIFAR-10 to test Munn et al. geometric complexity hypothesis.
- Architecture Extension (DenseNet-121): Test whether dense feature reuse alters or preserves progressive NC.
- Quantitative Statistical Correlation: Calculate Pearson & Spearman r values for NC metrics vs exit accuracy.
- Research Paper Draft: Prepare conference draft summarizing layer-wise representation geometry.

Thank You

Layer-wise Analysis of Feature Evolution in CNNs

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Questions & Discussion